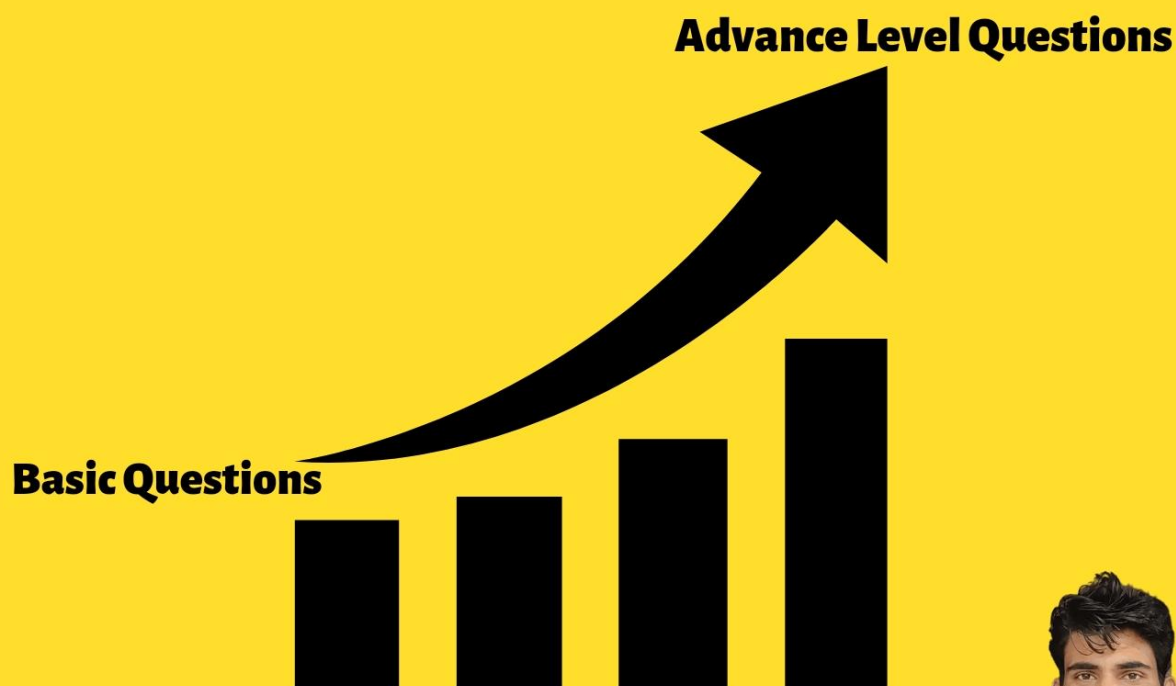


MATHS

HCF - NOTES

ALL TYPE WISE QUESTIONS



Maths with Gajanand



$\boxed{\text{HCF}} / \boxed{\text{GCD}} \rightarrow (\text{Greatest Common divisor})$

$\boxed{\text{Highest Common factor}}$

$$8 \Rightarrow \text{factor} = 1, 2, \textcircled{4}, 8$$

$$12 \Rightarrow \text{factor} = 1, 2, 3, \textcircled{4}, 6, 12$$

$$\text{HCF}(8, 12) = \boxed{4}$$

$\Rightarrow$ $\boxed{\text{HCF is the greatest Common factor of all given no.}}$

$$\text{HCF}(40, 64)$$

$$\begin{array}{cc} \downarrow & \downarrow \\ \textcircled{8} \times 5 & \textcircled{8} \times 8 \end{array} \quad \text{HCF} = 8$$

5 : 8
Coprime No. always

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$$\Rightarrow \text{HCF of two No.} = K$$

$$\text{Numbers} = Kx, Ky$$

$x, y \rightarrow \text{Coprime}$

$$\boxed{\text{LCM} = Kxy}$$

Ex. $\text{HCF}(a, b) = 3$

$$a : b = 4 : 5$$

$$\text{LCM} = 3 \times 4 \times 5$$

$$= \boxed{60} \text{ Ans}$$

Ex. HCF (48, 60)

$$\begin{array}{r} 48 \\ 60 \\ \hline \end{array}$$

diff = 12

HCF = 12

$\begin{array}{r} x \\ 4 \\ \hline \end{array}$ $\begin{array}{r} y \\ 5 \\ \hline \end{array}$

$\text{diff} = 1$

Ex. HCF (27, 45)

$$\begin{array}{r} 27 \\ 45 \\ \hline \end{array}$$

HCF = 9

$18 \rightarrow 9 \times 2$ $\therefore$ Both No. are divisible by 9

Ex. (323, 456, 703)

$$\begin{array}{r} 323 \\ 456 \\ 703 \\ \hline \end{array}$$

HCF = 19 all are divisible by 19

$133 = 19 \times 7$

③ Division method :-

HCF (902, 1517) = ??

$$\begin{array}{r} 902 \overline{) 1517} \quad (1 \\ - 902 \\ \hline 615 \end{array}$$

$$\begin{array}{r} 615 \overline{) 902} \quad (1 \\ - 615 \\ \hline 287 \end{array}$$

$$\begin{array}{r} 287 \overline{) 615} \quad (2 \\ - 574 \\ \hline 41 \end{array}$$

$$\begin{array}{r} 41 \overline{) 287} \quad (7 \\ - 287 \\ \hline 0 \end{array}$$

HCF $\rightarrow$ 41

HCF = 41 Ans

1, 1, 2, 7 $\rightarrow$ Successive Quotient

[1] In finding the HCF of two numbers by division method, the last divisor is 17 and the quotients are 1, 11 and 2, respectively. What is sum of the two numbers?

[SSC CGL Mains - 2019]

Soln:-

∴ Two Numbers are = 391 & 425

Sum = 816 Ans

$$\begin{array}{r}
 391 \overline{) 425} (1 \\
 \underline{-391} \\
 34 \\
 34 \overline{) 391} (11 \\
 \underline{-374} \\
 17 \\
 17 \overline{) 34} (2 \\
 \underline{-34} \\
 0
 \end{array}$$

[2] In finding the HCF of two numbers by division method four successive quotient are 4, 3, 6, and 5 respectively, and final divisor is 12. What are two numbers?

Soln:-

$$\begin{array}{r}
 1176 \overline{) 5076} (4 \\
 372 \\
 372 \overline{) 1176} (3 \\
 60 \\
 60 \overline{) 372} (6 \\
 12 \\
 12 \overline{) 60} (5 \\
 0
 \end{array}$$

Two No. = 1176, 5076 Ans

Type

* The greatest number that divides a, b, c and leaves r_1, r_2, r_3 as remainder respectively.

$$\text{No.} = \text{HCF}(a-r_1, b-r_2, c-r_3)$$

[3] Which greatest number will divides 3026 and 5053 leaving remainders 11 and 13 respectively?

Soln:- $\text{No.} = \text{HCF}(3026-11, 5053-13)$

$$= \text{HCF}(3015, 5040)$$

$$\ominus$$

$$2025 \Rightarrow 45 \times 45$$

$$\text{HCF} = \boxed{45} \text{ Ans}$$

[4] Find the greatest number that divides 797, 1085 and 1232 and leaves 16, 20, 25 as remainder respectively.

Soln:- $\text{No.} = \text{HCF}(797-16, 1085-20, 1232-25)$

$$= \text{HCF}(781, 1065, 1207)$$

$$\ominus$$

$$142 \Rightarrow (71) \times 2$$

$$\text{HCF} = \boxed{71} \text{ Ans}$$

[5] Find the greatest number that divides 556, 763 and 349 and leaves 4 as remainder respectively?

Soln:- $\text{No.} = \text{HCF}(556-4, 763-4, 349-4)$

$$= \text{HCF}(552, 759, 345)$$

$$\ominus$$

$$207 \Rightarrow 69 \times 3$$

$$\text{HCF} = \boxed{69} \text{ Ans}$$

Type-2

The greatest number that divides a, b, c and leaves the same remainder in each case.

$$[No. = HCF(a-b, b-c, a-c)]$$

- [6] Let x be the greatest number which when divides 955, 1027, 1075 the remainder in each case is the same. x is ?

Soln:-

$$x = HCF(1027-955, 1075-1027, 1075-955)$$

$$= HCF(72, 48, 120)$$

$$= \boxed{24} \text{ Ans}$$

- [7] When 7897, 8110 and 8536 are divided by the greatest number x , then the remainder in each is the same. The sum of the digit of x is :

Maths with Gajanand [SSC CGL Mains-2013]

Soln:-

$$x = HCF(8110-7897, 8536-8110, 8536-7897)$$

$$x = HCF(213, 426, 639)$$

$$x = 213$$

$$\text{Sum} = 2+1+3 = \boxed{6} \text{ Ans}$$

- [8] If r is the remainder when each of 6454, 7306 and 8797 is divided by the greatest number d then $d-r = ??$

Soln:- $No. = HCF(852, 1491, 2343)$

$$d = 213$$

$$R = \frac{6454}{213} = \text{rem. } 64$$

$$d-R = 213-64 = \boxed{149} \text{ Ans}$$

Mix Questions of LCM & HCF

⇒ Relation between LCM & HCF: For two No. a & b

$$* \quad a \times b = \text{Lcm}(a, b) \times \text{HCF}(a, b)$$

[9] The LCM of two numbers is 1820 and their HCF is 26.
If one number is 130 then the other number is:

Soln:-

$$130 \times x = 1820 \times 26$$
$$x = \boxed{364} \text{ Ans}$$

[10] The HCF of two number 12906 and 14818 is 478. Their LCM is:

Soln:-

$$12906 \times 14818 = 478 \times \text{LCM}$$
$$\text{LCM} = \frac{12906 \times 14818}{478} = \boxed{400086} \text{ Ans}$$

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[11] HCF and LCM of two numbers are 7 and 140 respectively.
If the numbers are between 20 and 45, the sum of the numbers is

Soln:-

$$a \times b = 7 \times 140$$
$$a \times b = 35 \times 28$$
$$a + b = 35 + 28 = \boxed{63} \text{ Ans}$$

[12] The HCF and LCM of two 2-digit numbers are 16 and 480 respectively. The number are:

Soln:-

$$a \times b = 16 \times 480$$
$$= 80 \times 96$$
$$a, b = 80, 96 \text{ Ans}$$

[13] The sum of and difference between the LCM and HCF of two numbers are 512 and 496, respectively. If one number is 72, then the other number is: [SSC CGL - 2022]

Soln:-

$$\text{LCM} = \frac{512 + 496}{2} = \frac{1008}{2}$$
$$= 504$$

$$\text{HCF} = \frac{512 - 496}{2} = \frac{16}{2}$$
$$= 8$$

$$a \times b = \text{LCM} \times \text{HCF}$$

$$72 \times b = 504 \times 8$$

$$b = \boxed{56} \text{ Ans}$$

[14] The HCF and LCM of two numbers are 44 and 264 respectively. If the first number is divided by 2, the quotient is 44. The other number is:

Soln:-

$$\text{1st No.} = 2 \times 44 = 88$$

$$88 \times x = 44 \times 264$$

$$x = \boxed{132} \text{ Ans}$$

[15] The sum of two numbers is 36 and their HCF and LCM are 3 and 105 respectively. The sum of the reciprocals of two numbers:

Soln:-

$$a + b = 36$$

$$a \times b = 3 \times 105$$

$$\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} = \frac{36}{3 \times 105} = \boxed{\frac{4}{35}} \text{ Ans}$$

[16] The LCM of two numbers is 44 times of their HCF. The sum of the LCM and HCF is 1125. If one number is 25, then the other number is:

Soln:- Let HCF = x
 LCM = $44x$
 $LCM + HCF = x + 44x = 1125$
 $45x = 1125$
 $x = 25$

HCF = 25

LCM = 25×44

$25 \times b = 25 \times 25 \times 44$ [$a \times b = LCM \times HCF$]
 $b = \boxed{1100}$ Ans

[17] The HCF of two numbers is 21 and their LCM is 221 times the HCF. If one of the numbers lies between 200 and 300, then the sum of the digits of other number is:

Soln:- HCF = 21
 LCM = 21×221
 $\quad \quad \quad \swarrow \searrow$
 $\quad \quad \quad 13 \times 17$

Ist No. = $21 \times 13 = \boxed{273}$

Digit sum

IInd No. = $21 \times 17 = \boxed{357} \Rightarrow 3 + 5 + 7 = \boxed{15}$

[18] The LCM of two numbers x and y is 204 times their HCF. If their HCF is 12 and the difference between the numbers is 60, then $x + y = ?$

Soln:-

HCF = 12

Let the no. $12x, 12y$

LCM = 204×12

$\quad \quad \quad \swarrow \searrow$
 $\quad \quad \quad 51 \times 4$
 $\quad \quad \quad \swarrow \searrow$
 $\quad \quad \quad 17 \times 3 \times 4$
 $= 17 \times 12$
 $\quad \quad \quad \underline{5}$

$12x - 12y = 60$

$x - y = 5$

Ist = 12×17) + = $12(17 + 12) = \boxed{348}$ Ans
 IInd = 12×12

[19] The HCF of two numbers is 29, and the other factors of their LCM are 15 and 13. The larger of the two numbers is: [SSC CPO-2020]

Soln:-

$$\text{HCF} = 29$$

$$\text{LCM} = 29 \times (15 \times 13) [\text{HCF} (a \times b)]$$

$$\text{Larger} = 29 \times 15 = \boxed{435} \text{ Ans}$$

$$\text{Small} = 29 \times 13 = \boxed{377}$$

[20] Two numbers are in the ratio 7:11. If their HCF is 28, then the sum of the two numbers is:

Soln:-

$$\text{HCF} = 28$$

$$a:b = 7:11$$

$$a+b = (7+11) \times 28$$

$$= 18 \times 28 = \boxed{504} \text{ Ans}$$

[21] The HCF of two numbers, each having three digits, is 17 and their LCM is 714. The sum of the numbers will be?

Soln:-

$$\text{HCF} = 17$$

$$17x, 17y$$

$$\text{LCM} = 714$$

$$17x \times 17y = 17 \times 714$$

$$x \times y = 42$$

$$\text{Possible value} = (1, 42), (2, 21), (3, 14), (6, 7)$$

$$\text{For three digit } (x, y) = (6, 7)$$

$$\text{No.} = \begin{array}{r} 102, 119 \\ \quad \cup \\ \quad + \end{array}$$

$$= \boxed{221} \text{ Ans}$$

22 Three numbers are in the ratio 2:3:4 and their HCF is 12. The LCM of the number is

Soln:-

$$\begin{aligned} \text{HCF} &= 12 \\ a:b:c &= 2:3:4 \Rightarrow 24, 36, 48 \\ \text{LCM} &= 12 \times (2 \times 3 \times 2) \\ &= \boxed{144} \text{ Ans} \end{aligned}$$

23 The HCF of two numbers is 8. which one of the following can never be their L.C.M.?

- (a) 24 (b) 48 (c) 56 ☒ (d) 60

Soln:- LCM. will be multiple of HCF=8 only (d) is not multiple of 8

24 The LCM of three different numbers is 120. Which of the following cannot be their HCF?

- Soln:- (a) 8 (b) 12 (c) 24 ☒ (d) 35

LCM is the product of one common factor and other different factors of the given numbers

$$\begin{aligned} \text{LCM} &= 120 = 2 \times 2 \times 3 \times 5 \times 2 \\ &= 4 (3 \times 5 \times 2) \end{aligned}$$

Common factor

so HCF will be multiple of 4

25 Three numbers are in the proportion of 3:8:15 and their LCM is 8280. What is their HCF?

- (a) 60 ☒ (b) 23 (c) 76 (d) 57 [SSC CGL-2022]

Soln:- let No. are = 3k, 8k, 15k

$$\text{LCM} = 3 \times 8 \times 15k = 8280$$

$$k = 23$$

HCF will be 23 either multiple of 23

26 The HCF of a, b is 12, a, b are positive integers and $a > b > 12$. The smallest values of (a, b) are respectively.
 (a) 12, 24 (b) 24, 12 (c) 24, 36 ☒ (d) 36, 24

Soln:- $a > b > 12$
 so 36, 24 are smallest value.

27 Sum of two numbers is 384. HCF of the numbers is 48. The difference of the numbers is:

Soln:- HCF = 48
 No. = $48x, 48y$
 $48(x+y) = 384$
 $x+y = 8$
 $(x, y) = (1, 7) (3, 5)$
 $= (48, 336), (144, 240)$
 $x-y = \boxed{288} \quad \boxed{96}$

28 The product of the LCM and HCF of two numbers is 24. The difference of the two numbers is 2. Find the numbers?
 (a) 8 and 6 (b) 8 and 10 (c) 2 and 4 (d) 6 and 4

Soln:- LCM $\times$ HCF = 24
 so $a \times b = 24$
 $a - b = 2$
 $(a, b) = (4, 6)$ Ans

29 The ratio of the sum to the LCM of two natural numbers is 7:12. If their HCF is 4, then the smaller number is:

Soln:- HCF = 4, No. = $4x, 4y$
 LCM = $4xy$
 $\frac{4x+4y}{4xy} = \frac{7}{12}$
 $12(x+y) = 7xy$
 $x = 3, y = 4$
 Smaller = $3 \times 4 = \boxed{12}$ Ans

30. The LCM of two positive integers is twice the larger number. The difference of the smaller number and the HCF of the two numbers is 4. The smaller number is:

Soln:- Let HCF = a

$$\text{No.} = ax, ay \quad (ax > ay)$$

$$\text{LCM} = 2ax$$

$$axy = 2ax$$

$$\boxed{y = 2}$$

$$\text{Smaller No.} - \text{HCF} = 4$$

$$ay - a = 4$$

$$2a - a = 4$$

$$\boxed{a = 4}$$

$$\text{Smaller No.} = ay = 4 \times 2$$

3

$$= \boxed{8} \text{ Ans}$$

31. If the HCF and LCM of two consecutive (positive) even numbers be 2 and 84 respectively, then the sum of the numbers is:

Soln:- HCF = 2, No. = $2x, 2y$

$$2xy = 84$$

$$xy = 42$$

$$\begin{matrix} \wedge \\ 7 \times 6 \end{matrix}$$

$$\text{No.} = 14, 12 \Rightarrow \overset{\text{Sum}}{14 + 12} = \boxed{26} \text{ Ans}$$

32. If $P = 2^3 \times 3^{10} \times 5$, $Q = 2^5 \times 3 \times 7$, then HCF of P and Q is:

Soln:- $P = 2^3 \times 3^{10} \times 5$

$$Q = 2^5 \times 3 \times 7$$

HCF = minimum common power

$$= 2^3 \times 3 \text{ Ans}$$

[33] If A and B are the HCF and LCM respectively of two algebraic expressions x and y, and $A+B = x+y$, then the value of A^3+B^3 is :

- (a) x^3-y^3 (b) x^3 (c) y^3 (d) x^3+y^3

Soln:-

$$\text{LCM} = B$$

$$\text{HCF} = A$$

$$A \times B = \text{LCM} \times \text{HCF}$$

$$x \times y = AB \quad \text{--- (1)}$$

$$x+y = A+B \quad \text{--- (2)}$$

$$\begin{aligned} A^3+B^3 &= (A+B)^3 - 3AB(A+B) \\ &= (x+y)^3 - 3xy(x+y) \\ &= (x^3+y^3) \quad \underline{\underline{\text{Ans}}} \end{aligned}$$

[34] Three numbers are in the ratio $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$. If the difference between the greatest number and the smallest number is 33, the HCF of the three numbers is :

[SSC CGL Mains-2022]

Soln:-

$$\left(\frac{1}{2} : \frac{2}{3} : \frac{3}{4} \right) \times 12$$

$$\Rightarrow \begin{matrix} \text{I} & \text{II} & \text{III} \\ 6 & 8 & 9 \end{matrix}$$

$$\underbrace{\quad \quad \quad}_{\text{3 Unit}}$$

$$3 \text{ Unit} \longrightarrow 33$$

$$1 \text{ Unit} \longrightarrow 11$$

$$\text{I} = 11 \times 6 = 66$$

$$\text{II} = 11 \times 8 = 88$$

$$\text{III} = 11 \times 9 = 99$$

$$\text{HCF}(66, 88, 99) = \underline{\underline{11}} \quad \underline{\underline{\text{Ans}}}$$

[35] 13, a, b, c are four distinct numbers and the HCF of each pair of numbers (13, a); (13, b); (13, c) is 13, where a, b, c are each less than 60 and $a < b < c$. What is the value of $\frac{a+c}{b}$?

Soln:- $(13, a), (13, b), (13, c) = \text{HCF} = 13$
 $\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 $26 \quad < \quad 39 \quad < \quad 52$ $a < b < c$

$$\frac{a+c}{b} = \frac{26+52}{39} = \boxed{2} \text{ Ans}$$

[36] The sum of two positive numbers is 240 and their HCF is 15. Find the number of pairs of numbers satisfying the given condition. [SSC CGL - 2020]

Soln:- $\text{HCF} = 15$
 $\text{No.} = 15x, 15y$
 $15x + 15y = 240$
 $x + y = 16$

$(1, 15), (3, 13), (5, 11), (7, 9)$ $\textcircled{4} \text{ Ans}$

[37] The HCF and product of two numbers are 15 and 6300 respectively. The number of possible pairs of the number is.

Soln:- $\text{HCF} = 15$
 $\text{No.} = 15x, 15y$
 $15x \times 15y = 6300$
 $xy = 28$

$(1, 28), (7, 4)$ Total Pairs
 $\boxed{2} \text{ Ans}$

[38] The ratio of HCF and LCM of two numbers a and b is $1 : 30$ and the difference between the HCF and LCM is 493 . Find the possible number of pairs of a and b ?

Soln:-

$$\text{HCF} : \text{LCM}$$

1 : 30

$$29 \rightarrow 493$$

1 Unit $\rightarrow$ 17

$$\text{HCF} = 17 \quad \rightarrow \text{No.} = 17x, 17y$$
$$\text{LCM} = 17 \times 30 = 510$$

$$17x \times 17y = 17 \times 510$$

$$xy = 30$$

$$(1, 30), (2, 15), (3, 10), (5, 6)$$

Pairs = $\boxed{4}$ Ans

39 The HCF of $x^6 - 1$ and $x^9 + 2x^3 - 2x - 1$ is:

(9) $x^2 + 1$

(b) $x-1$

~~(e)~~ $x^2 - 1$

(d) $x + 1$

Soln:- $x^6 - 1 = (x^2 - 1)(x^4 + 1 + x^2)$ — (1)

$$\therefore a^3 - b^3 = (a-b)(a^2 + b^2 + ab)$$

$$x^4 + 2x^3 - 2x - 1$$

$$x^4 - 1 + 2x(x^2 - 1)$$

$$(x^2-1)(x^2+1) + 2x(x^2-1)$$

$$(x^2-1) [x^2+1+2x] \text{ --- (2)}$$

From ① & ② Common term = HCF = $x^2 - 1$ Ans

LCM & HCF of fractions

$$\frac{a}{b}, \frac{c}{d}, \frac{e}{f}$$

$$\left[\text{LCM} = \frac{\text{LCM of Numerator}}{\text{HCF of Denominator}} \right]$$

$$= \frac{\text{Lcm}(a, c, e)}{\text{Hcf}(b, d, f)}$$

$$\left[\text{HCF} = \frac{\text{HCF of Numerator}}{\text{LCM of Denominator}} \right]$$

$$= \frac{\text{Hcf}(a, c, e)}{\text{Lcm}(b, d, f)}$$

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40 HCF of $\frac{2}{3}$, $\frac{4}{5}$ and $\frac{6}{7}$

Soln:- $\text{HCF} = \frac{\text{HCF}(2, 4, 6)}{\text{LCM}(3, 5, 7)} = \frac{2}{105} \text{ Ans}$

41 LCM of $\frac{64}{45}$, $\frac{16}{15}$, $\frac{32}{35}$, $\frac{48}{25}$

Soln:- $\text{LCM} = \frac{\text{LCM}(64, 16, 32, 48)}{\text{HCF}(45, 15, 35, 25)}$
 $= \frac{192}{5} \text{ Ans}$

41 Find HCF and LCM of the following fraction?

(a) 0.9, 2.7, 0.36

(b) $\frac{4}{5}$, $\frac{6}{8}$, $\frac{8}{25}$

Soln:-
(a) $\frac{9}{10}$, $\frac{27}{10}$, $\frac{36}{100}$ ← make same

$$\frac{90}{100}, \frac{270}{100}, \frac{36}{100}$$

$$\begin{aligned} \text{LCM} &= \frac{\text{LCM}(90, 270, 36)}{\text{HCF}(100, 100, 100)} \\ &= \frac{540}{100} = \boxed{5.4} \end{aligned}$$

$$\begin{aligned} \text{HCF} &= \frac{\text{HCF}(90, 270, 36)}{\text{LCM}(100, 100, 100)} \\ &= \frac{18}{100} = \boxed{0.18} \end{aligned}$$

(b) $\frac{4}{5}$, $\frac{6}{8}$, $\frac{8}{25}$

$$\begin{aligned} \text{LCM} &= \frac{\text{LCM}(4, 6, 8)}{\text{HCF}(5, 8, 25)} \\ &= \frac{24}{1} = \boxed{24} \text{ Ans} \end{aligned}$$

$$\begin{aligned} \text{HCF} &= \frac{\text{HCF}(4, 6, 8)}{\text{LCM}(5, 8, 25)} \\ &= \frac{2}{200} \\ &= \boxed{\frac{1}{100}} \text{ Ans} \end{aligned}$$

42 HCF of $(13^7 + 11^7)$ and $(13^5 + 11^5)$

Soln

$a^n + b^n \Rightarrow$ always divisible by $(a+b)$

if n is odd

$$\text{HCF} = (a+b)$$

$$= 13+11$$

$$= \boxed{24} \text{ Ans}$$

$$\Rightarrow [a^n \pm 1, b^m \pm 1] = \boxed{a^{\text{HCF}(n,m)} \pm 1}$$

$n, m \rightarrow \text{odd}$

Ex. $4^{63} + 1$ & $4^{45} + 1$ HCF = ??

Soln:- m-I

$$4^{63} + 1 = (4^9)^7 + 1^7$$

$$4^{45} + 1 = (4^9)^5 + 1^5$$

$$\text{HCF} = a + b$$

$$= \boxed{4^9 + 1} \text{ Ans}$$

m-II

$$\text{HCF} = a^{\text{HCF}(n,m)} \pm 1$$

$$= 4^{\text{HCF}(63,45)} + 1$$

$$= \boxed{4^9 + 1} \text{ Ans}$$

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Ex. $2^{25} - 1$ and $4^{65} - 1$

$$\text{HCF} = 2^{\text{HCF}(25,65)} - 1 = 2^5 - 1 = \boxed{31} \text{ Ans}$$

Ex. $3^{333} + 1$ and $3^{334} + 1$ HCF = ??

Soln:-

$$= 3^{\text{HCF}(333, 334)} + 1$$

$$= \boxed{3^{333} + 1} \text{ Ans}$$

[43] Three tankers contain 403 litres, 434 litres, 465 litres of diesel respectively. Then the maximum capacity of a container that can measure the diesel of the three containers exact number of times is.

Soln:- $\text{HCF}(403, 434, 465)$
 $\text{HCF} = \boxed{31}$ all No. are divisible by 31

[44] There are 24 peaches, 36 apricots and 60 bananas. and they have to be arranged in several rows in such a way that every row contains the same number of fruits of only one type. What is the minimum number of row required for this to happen?

Soln:- m-I $\text{HCF}(24, 36, 60) = 12$
 $\text{No. of Rows} = \frac{24}{12} + \frac{36}{12} + \frac{60}{12}$
 $= 2 + 3 + 5 = \boxed{10} \text{ Ans}$

m-II

12	24, 36, 60
	2, 3, 5

← should be coprime No.
 $\oplus$
 $\boxed{10} \text{ Ans}$

[45] Two pipes of length 1.5m and 1.2m are to be cut into equal pieces without leaving any extra length of Pipes. The greatest length of the pipe pieces of same size which can be cut from these two lengths will be:

Soln:- $\text{HCF}(1.5, 1.2) = \boxed{0.3} \text{ Ans}$

[46] In a school, 391 boys and 323 girls have been divided into the largest possible equal classes, so that each class of boys numbers the same as each class of girls, What is the number of classes?

Soln:- HCF (391, 323)

$$\begin{array}{c} \textcircled{68} \\ 68 \\ \text{17} \times 4 \end{array}$$

HCF = $\boxed{17}$ which divides the given no.

[47] A farmer has 945 cows and 2475 sheep. He farms them into flocks, keeping cows and sheep separate and having the same number of animals in each flock. If these flocks are as large as possible, then the maximum number of animals in each flock and total number of flocks required for the purpose are respectively.

Soln:-

$$\text{HCF} = \left\{ \begin{array}{c|c} 5 & 945, 2475 \\ \hline 9 & 189, 495 \\ \hline 21 & 55 \end{array} \right.$$

$\uparrow$ $\uparrow$
 Cows Sheep

maximum animal in each flock
HCF = $\boxed{45}$

[48] There are three farms, whose area is 288, 408 and 552 square meters, respectively. In which rows of equal length has to be made. If the width of each row is 4 meters, then what will be the maximum length?

Soln:- HCF (288, 408, 552)

$$\begin{array}{c} \textcircled{120} \\ 120 \\ \text{5} \times \textcircled{24} \end{array}$$

HCF = $\boxed{24}$



$$l \times 4 = 24$$

$$\boxed{l = 6m} \quad \underline{\text{Ans}}$$